MIL-C-11272/6A
7 January 1970
SUPERSEDING
MIL-C-11272/6
10 August 1960

MILITARY SPECIFICATION

# CAPACITORS, FIXED, GLASS DIELECTRIC,

# STYLE CY13

*This specification is mandatory for use by all Departments and Agencies of the Department of Defense.*

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-C-23269.

## 1. SCOPE

1.1 This specification covers the detail requirements for style CY13, axial-lead, glass-dielectric (vitreous enamel), fixed capacitors.

## 2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-C-11272 Capacitors, Fixed, Glass Dielectric, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

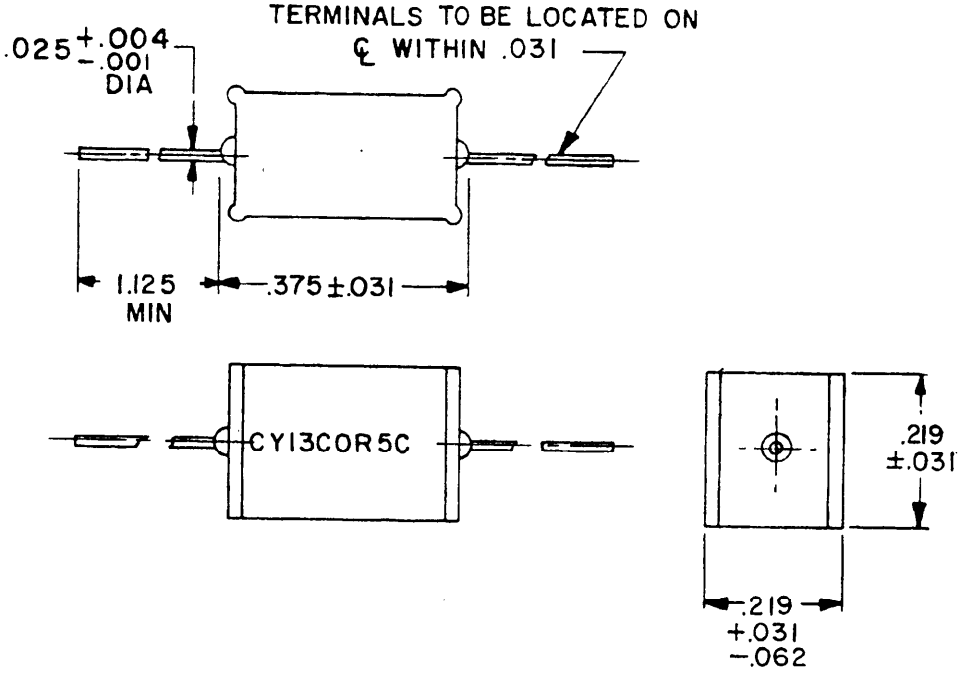
## 3. REQUIREMENTS

3.1 **Requirements.** Requirements shall be in accordance with MIL-C-11272, and as specified herein.

3.2 **Design and construction.** Capacitors shall be of the design, construction, and physical dimension specified on figure and in table I.



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| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .001 | .03 | .062 | 1.57 |
| .004 | .10 | .219 | 5.56 |
| .025 | .64 | .375 | 9.53 |
| .031 | .79 | 1.125 | 28.58 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Style CY13 capacitors.



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TABLE I. *Style CY13.*

| Type designation[1/] | Capacitance | DC rated voltage | Capacitance tolerances available | Type designation[1/] | Capacitance | DC rated voltage | Capacitance tolerances available |
|---|---|---|---|---|---|---|---|
| | *pF* | *volts* | *(±)* | | *pF* | *volts* | *(±)* |
| CY13C0R5C | 0.5 | 500 | C | CY13C240- | 24 | 500 | GJ |
| CY13C1R0C | 1.0 | 500 | C | CY13C270- | 27 | 500 | FGJ |
| CY13C1R5C | 1.5 | 500 | C | CY13C300- | 30 | 500 | FGJ |
| CY13C2R2- | 2.2 | 500 | CD | CY13C330- | 33 | 500 | FGJ |
| CY13C2R7C | 2.7 | 500 | C | CY13C360- | 36 | 500 | FGJ |
| CY13C3R0- | 3.0 | 500 | CD | CY13C390- | 39 | 500 | FGJ |
| CY13C3R3C | 3.3 | 500 | C | CY13C430- | 43 | 500 | FGJ |
| CY13C3R6- | 3.6 | 500 | CD | CY13C470- | 47 | 500 | FGJ |
| CY13C3R9C | 3.9 | 500 | C | CY13C510- | 51 | 500 | FGJ |
| CY13C4R3- | 4.3 | 500 | CD | CY13C560- | 56 | 500 | FGJ |
| CY13C4R7C | 4.7 | 500 | C | CY13C620- | 62 | 500 | FGJ |
| CY13C5R1C | 5.1 | 500 | C | CY13C680- | 68 | 500 | FGJ |
| CY13C5R6- | 5.6 | 500 | CJ | CY13C750- | 75 | 500 | FGJ |
| CY13C6R2- | 6.2 | 500 | CJ | CY13C820- | 82 | 500 | FGJ |
| CY13C6R8- | 6.8 | 500 | CJ | CY13C910- | 91 | 500 | FGJ |
| CY13C7R5- | 7.5 | 500 | CJ | CY13C101- | 100 | 500 | FGJ |
| CY13C8R2- | 8.2 | 500 | CJ | CY13C111- | 110 | 500 | FGJ |
| CY13C9R1- | 9.1 | 500 | CJ | CY13C121- | 120 | 500 | FGJ |
| CY13C100- | 10 | 500 | CJ | CY13C131- | 130 | 500 | FGJ |
| CY13C110- | 11 | 500 | CJ | CY13C151- | 150 | 500 | FGJ |
| CY13C120- | 12 | 500 | CJ | CY13C161- | 160 | 500 | FGJ |
| CY13C130- | 13 | 500 | GJ | CY13C181- | 180 | 500 | FGJ |
| CY13C150- | 15 | 500 | GJ | CY13C201- | 200 | 500 | FGJ |
| CY13C160- | 16 | 500 | GJ | CY13C221- | 220 | 300 | FGJ |
| CY13C180- | 18 | 500 | GJ | CY13C241- | 240 | 300 | FGJ |
| CY13C200- | 20 | 500 | GJ | CY13C271- | 270 | 300 | FGJ |
| CY13C220- | 22 | 500 | GJ | CY13C301- | 300 | 300 | FGJ |

[1/] Complete type designation will include an additional letter symbol to indicate capacitance tolerance, where applicable.

**3.3 Life**

3.3.1 *For qualification inspection.* Capacitors shall meet the following requirements, as applicable:

(a) At 25°C:
  (1) Insulation resistance – 100,000 megohms or 100 ohmfarads, whichever is less.
  (2) Dissipation factor – Shall not exceed 0.002.

(b) At 125°C:
  (1) Insulation resistance – 10,000 megohms or 10 ohmfarads, whichever is less.
  (2) Dissipation factor – Shall not exceed 0.003.

3.3.2 *For quality conformance inspection.*



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3.3.2.1 *Performance check.* Capacitors shall meet the following requirements:

(a) Insulation resistance – 100,000 megohms or 100 ohmfarads, whichever is less.
(b) Dissipation factor – Shall not exceed 0.002.

3.4 **Quality factor (Q).** The Q shall be not less than 2,000.

3.5 **Temperature coefficient and capacitance drift.** The temperature coefficient and capacitance drift shall be within the limits specified in table II.

TABLE II. *Temperature coefficient and capacitance drift.*

| Temperature coefficient | Capacitance drift (-55° to +125°C) |
|---|---|
| *Parts/million/°C* 105 ± 25 | 0.1 percent or 0.1 pF, whichever is greater. |

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-C-11272.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-11272.

6. NOTES

6.1 **Notes.** In addition to the notes specified herein, the notes specified in MIL-C-11272 are applicable to this specification.

Custodians:
Army – EL
Navy – EC
Air Force – 11

Review activities:
Army – EL
Navy – EC, SH
Air Force – 11, 17, 85
DSA – ES

User activities:
Army –
Navy –
Air Force – 19

Preparing activity:
Air Force – 11

Agent:
DSA – ES

(Project 5910-0994-7)



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